

changes in the intervertebral discs and associated structural abnormalities, including osteophyte formation, ligament hypertrophy or calcification, and facet joint dysfunction.^{5,6} Although these structural alterations typically develop gradually over years or decades, patients often seek medical attention during acute or subacute exacerbations triggered by mechanical loading, sustained postural stress, or minor trauma. CS-related neck pain imposes a substantial socio-economic and psychological burden.^{7–10}

Conservative management remains the mainstay of treatment for CS-related neck pain.^{11,12} Current therapeutic approaches primarily include pharmacological interventions, particularly non-steroidal anti-inflammatory drugs (NSAIDs), along with physical therapy and structured rehabilitation programs.^{13,14} Although NSAIDs are commonly prescribed for symptom relief, their prolonged use is associated with potential gastrointestinal, cardiovascular, and other adverse effects, which may restrict their long-term clinical application.^{15–19} Consequently, there is an increasing clinical need to explore safer and more effective therapeutic strategies for the management of CS-related neck pain.

Acupotomy is a minimally invasive intervention that combines acupuncture principles with targeted soft-tissue release techniques.^{20–22} Its flat-bladed tip is designed to facilitate both acupoint stimulation and mechanical dissection of adhesions, which may help reduce local tissue tension and alleviate related functional limitations.^{23,24} With the advancement of ultrasound-guided techniques, real-time visualization of anatomical structures may further improve procedural accuracy and safety during acupotomy procedures. Previous clinical and experimental studies have suggested that an ultrasound-guided acupotomy-based treatment strategy may alleviate musculoskeletal pain and improve patients' quality of life.^{25–27} In addition, our preliminary animal studies suggest that acupotomy may indirectly influence intervertebral disc homeostasis by alleviating paraspinal soft-tissue tension and improving local biomechanical loading conditions. These preclinical findings provide a biological rationale for further clinical investigation of acupotomy-based interventions in spinal degenerative disorders. However, this indirect mechanistic linkage requires further validation through dedicated translational studies.^{28,29} Importantly, despite these promising findings, methodologically rigorous evidence from randomized controlled trials evaluating an ultrasound-guided acupotomy-based treatment strategy for CS-related neck pain remains limited, particularly in comparison with commonly used first-line NSAID therapy. Previous studies have been constrained by methodological limitations, including non-randomized designs, small sample sizes, heterogeneous comparator interventions, and inadequate blinding, which may limit the interpretation and generalizability of their findings. Therefore, the present study is designed as a prospective, randomized, assessor- and statistician-blinded parallel-group controlled trial to evaluate the comparative effectiveness and safety of an ultrasound-guided acupotomy-based treatment strategy compared with celecoxib in patients with CS-related neck pain. The primary hypothesis is that this treatment strategy may provide clinically meaningful improvements in pain intensity and functional outcomes compared with NSAID therapy. This trial incorporates several methodological features, including randomized allocation, assessor and statistician blinding, a standardized NSAID comparator, and sample size determination based on statistical power analysis, to enhance the methodological quality and clinical interpretability of the planned evaluation.

Material and Methods

Study Design

This study is designed as a prospective, randomized, assessor- and statistician-blinded, parallel-group clinical trial. A total of 104 participants will be randomly assigned in a 1:1 ratio to either the acupotomy group or the control group. The study protocol was developed in accordance with the SPIRIT guidelines and will be reported following the CONSORT statement.^{30,31} Figure 1 presents a flow chart of the study design.

Participants

Participants will be recruited from The First Affiliated Hospital of Shandong First Medical University and Shandong Provincial Qianfoshan Hospital. Eligible participants will be screened according to predefined inclusion and exclusion criteria and will provide written informed consent prior to enrollment.

Inclusion Criteria

Participants must meet all of the following criteria: